HEART DISEASE PREDICTION USING LOGISTIC REGRESSION, SVM, AND K-NEAREST NEIGHBORS

Machine Learning · Bachelor of Technology, CSE

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ABSTRACT

Heart disease is one of the major health concerns worldwide. This project develops a Machine Learning-based system for predicting heart disease using Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The system uses a heart disease dataset containing 1,025 patient records with 13 features. The data is preprocessed and divided into training and testing sets before applying the machine learning algorithms. The models are evaluated using different performance metrics to study their prediction performance. The project demonstrates how Machine Learning can be applied to patient-related data for early risk identification.

INTRODUCTION

- Heart disease is a major global health concern.
- Early identification can support timely treatment.
- Patient data contains useful information for prediction.
- Machine Learning can identify patterns in patient data.
- This project predicts whether heart disease is present or not.

PROBLEM STATEMENT

CURRENT CHALLENGES

- Diagnosing heart disease can require time and medical expertise.
- Large amounts of patient data can be difficult to analyze manually.
- Identifying useful patterns from patient attributes is challenging.
- A Machine Learning-based prediction system can help analyze the data.

PROJECT GOAL

To develop a Machine Learning system that predicts the likelihood of heart disease using patient-related attributes.

DATASET

Attribute	Details
Dataset	Heart Disease Dataset
Source	Kaggle
Number of Records	1,025
Number of Features	13
Target Variable	Presence or absence of heart disease

Important Features: Age · Sex · Chest Pain Type · Resting Blood Pressure · Cholesterol · Maximum Heart Rate · Exercise-Induced Angina · ST Depression

ALGORITHMS USED

1. LOGISTIC REGRESSION

Used for classification and predicts the probability of heart disease.

2. SUPPORT VECTOR MACHINE (SVM)

A supervised learning algorithm used for classification.

3. K-NEAREST NEIGHBORS (KNN)

Predicts the class based on nearby data points.

WORKFLOW

01

HEART DISEASE DATASET

02

DATA PREPROCESSING

03

TRAIN-TEST SPLIT

04

MACHINE LEARNING MODELS

Logistic Regression · SVM · KNN

05

PREDICTION

06

MODEL EVALUATION

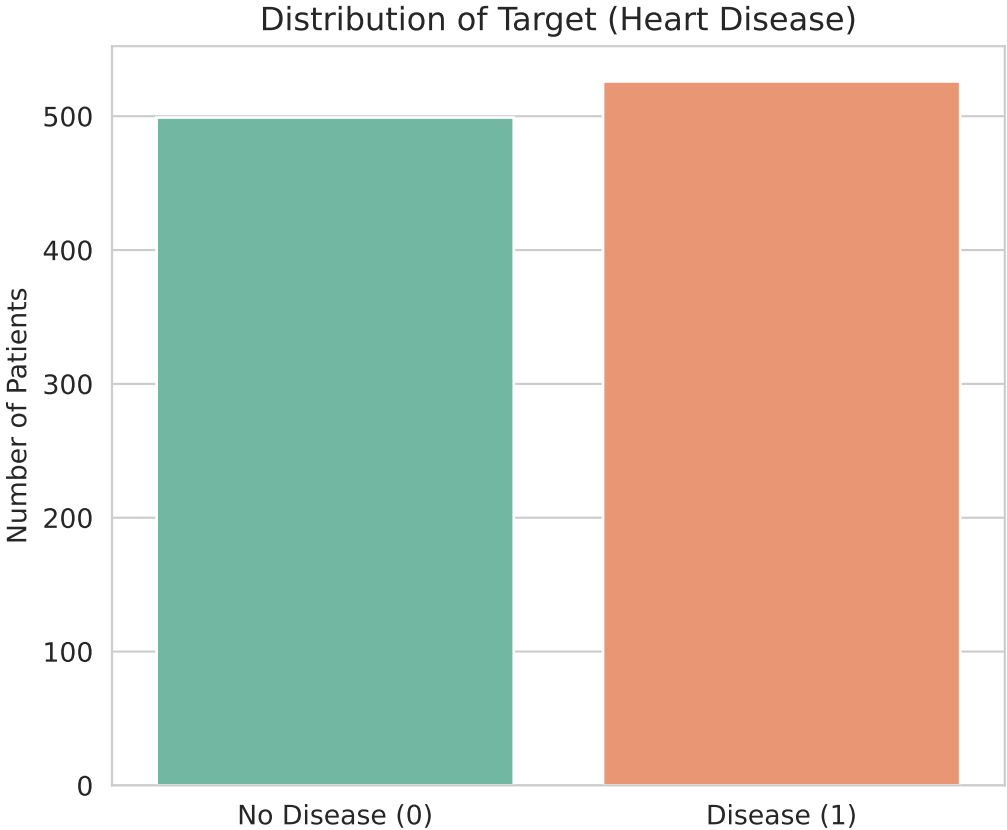
Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC

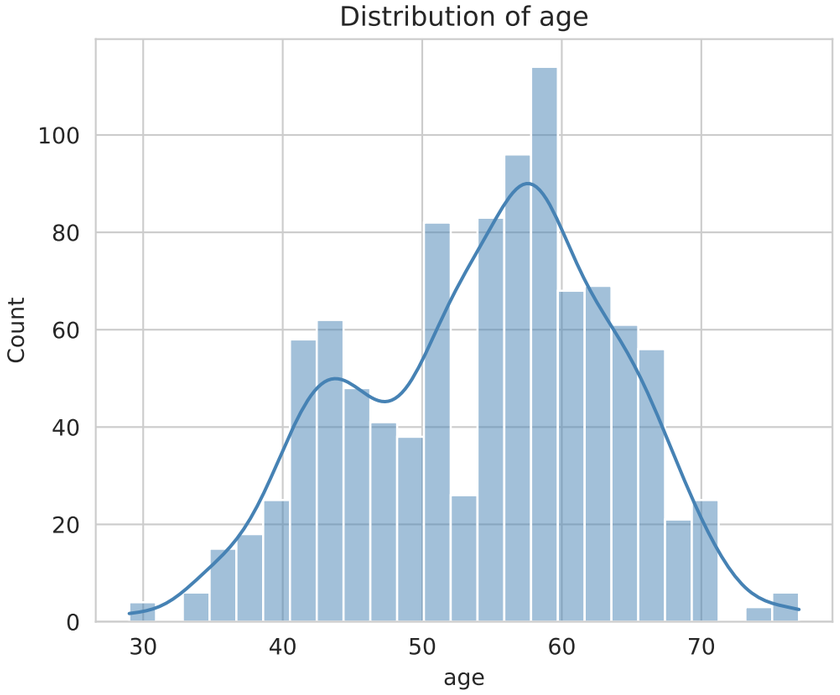
CONCLUSION

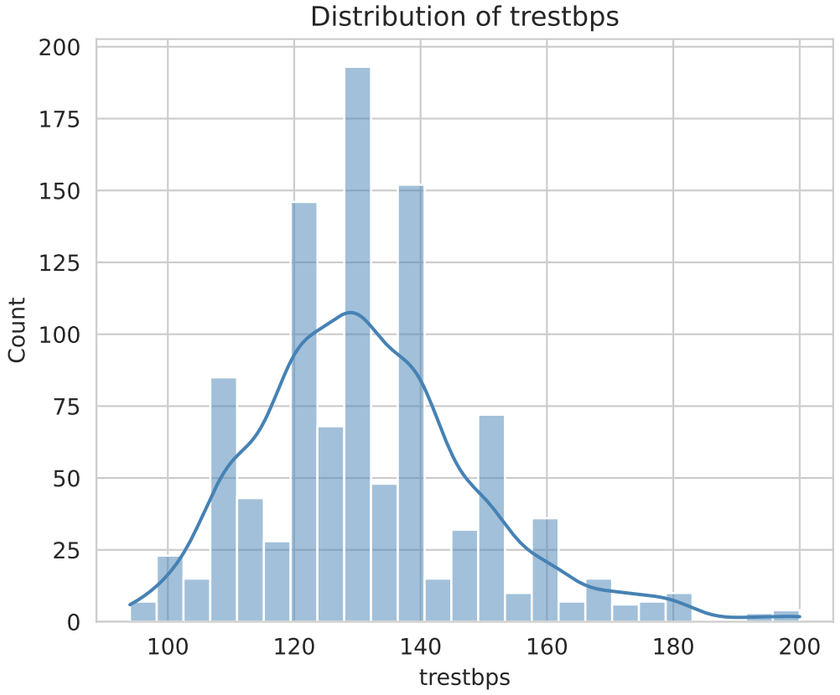
- The project applies Machine Learning to heart disease prediction.
- Logistic Regression, SVM, and KNN are used for classification.
- Patient-related features are used to make predictions.
- The models are evaluated using different performance metrics.
- The system demonstrates the practical use of Machine Learning in healthcare.

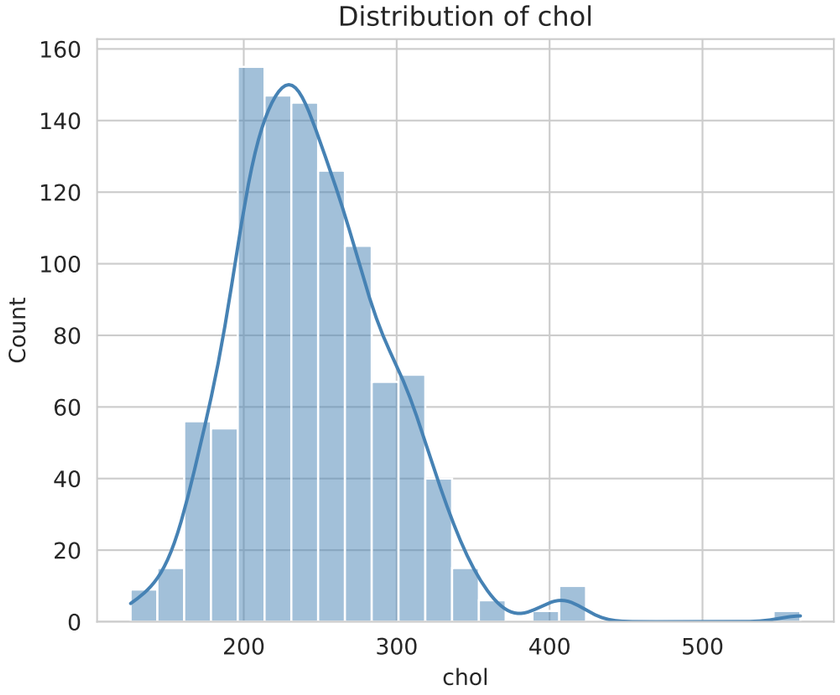
Machine Learning can support early risk identification, while medical diagnosis should always be performed by qualified healthcare professionals.

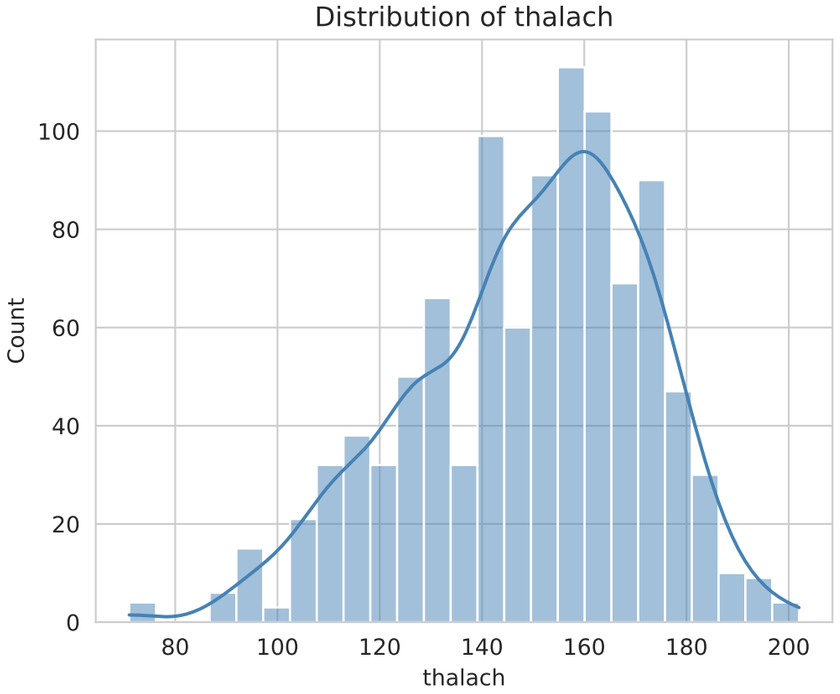

Missing Value Heatmap
0
38
76
114
152
190
228
266
304
342
380
418
456
494
532
570
608
646
684
722
760
798
836
874
912
950
988
age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target

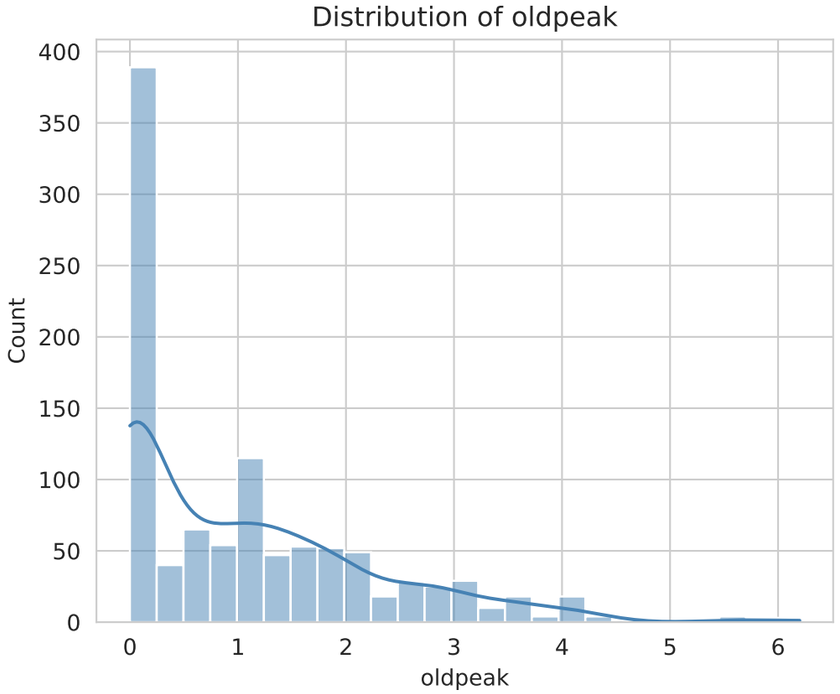

Distribution of Target (Heart Disease)
Number of Patients
0
100
200
300
400
500
No Disease (0)
Disease (1)

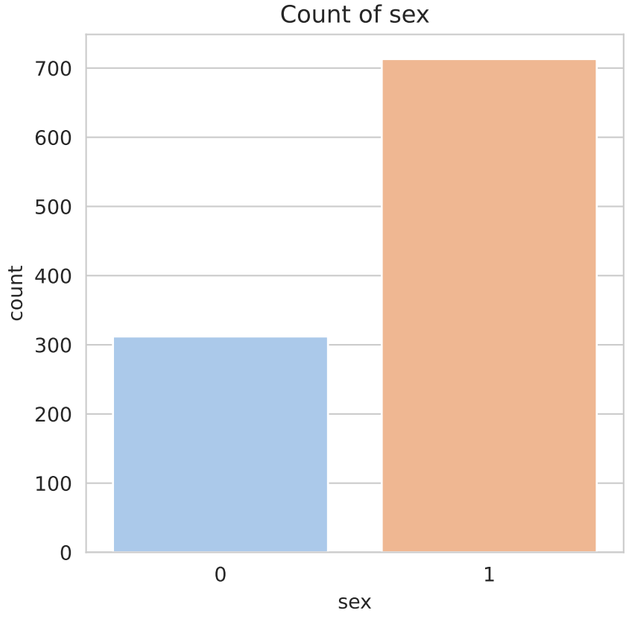

Distribution of age
Count
age
0
20
40
60
80
100
30
40
50
60
70

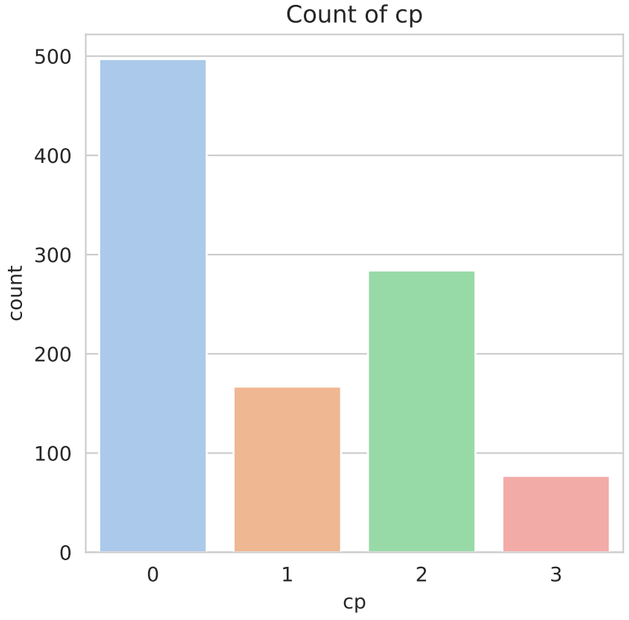

Distribution of trestbps
Count
trestbps
0
25
50
75
100
125
150
175
200
100
120
140
160
180
200

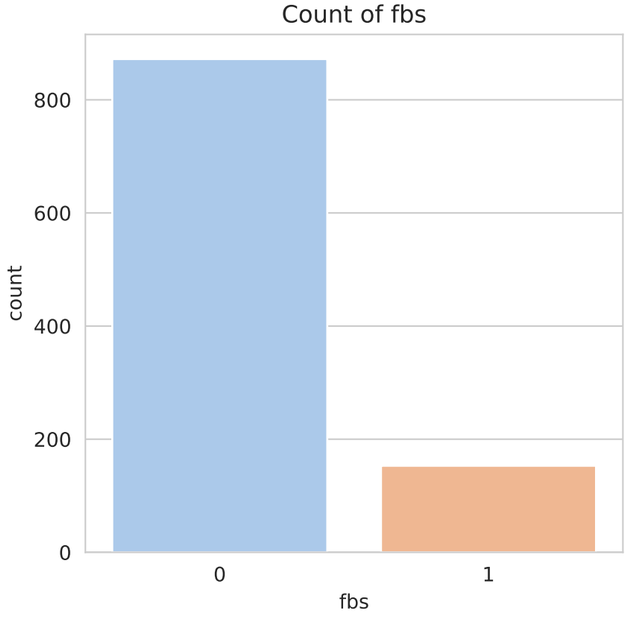

Distribution of chol
Count
chol
0
20
40
60
80
100
120
140
160
200
300
400
500

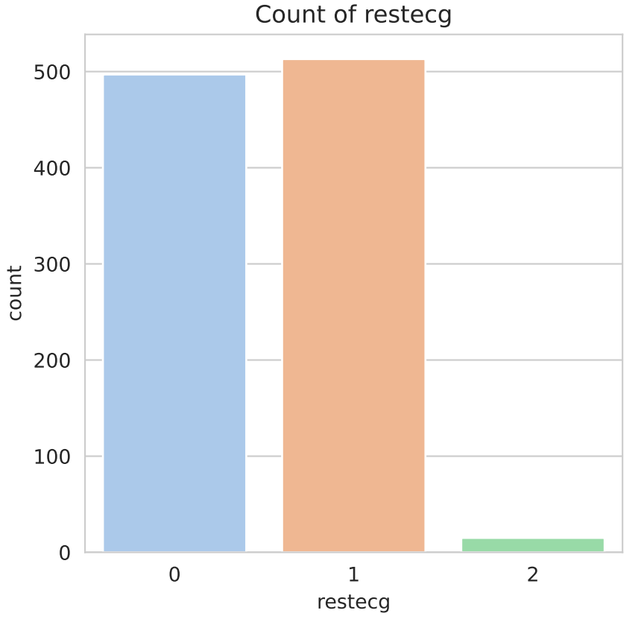

Distribution of thalach
Count
thalach
0
20
40
60
80
100
80
100
120
140
160
180
200

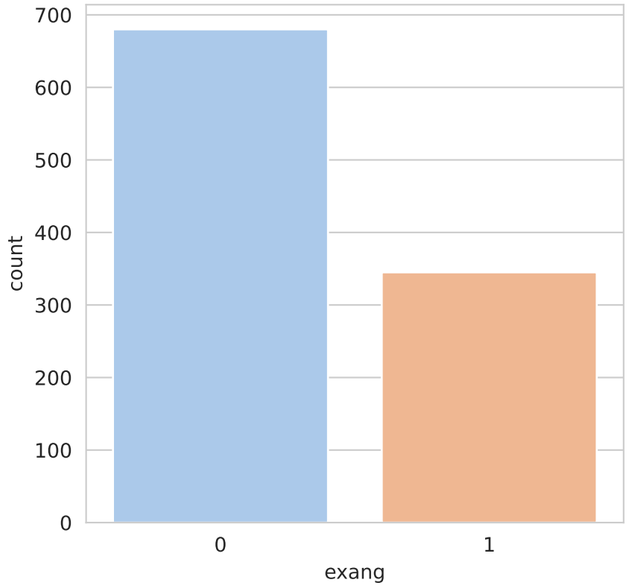

Distribution of oldpeak
Count
oldpeak
0
50
100
150
200
250
300
350
400
0
1
2
3
4
5
6

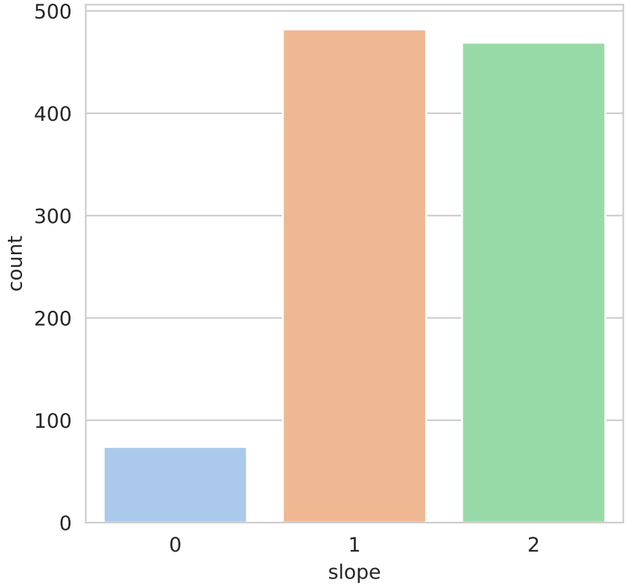

Count of sex
count
sex
0
1

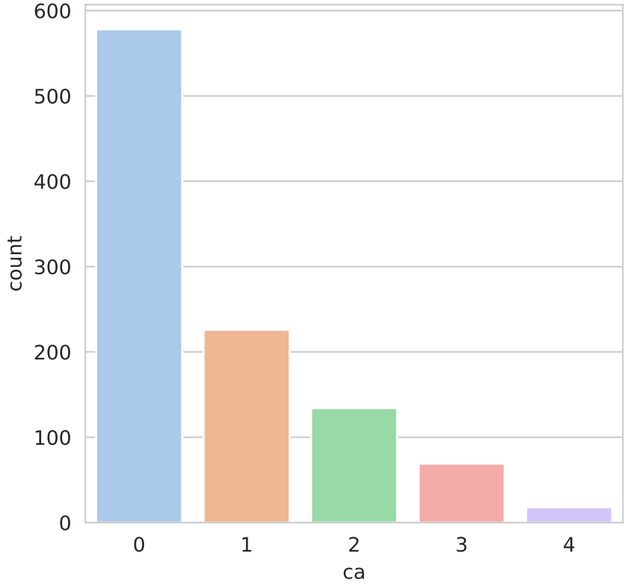

Count of cp
count
cp
0
1
2
3

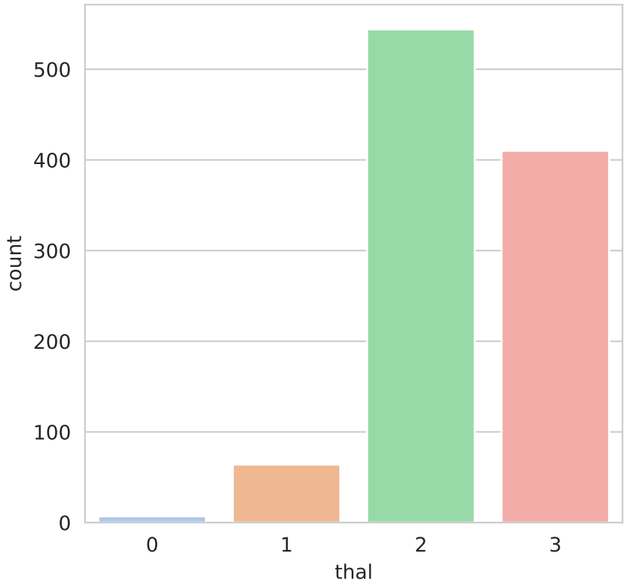

Count of fbs
count
fbs
0
1

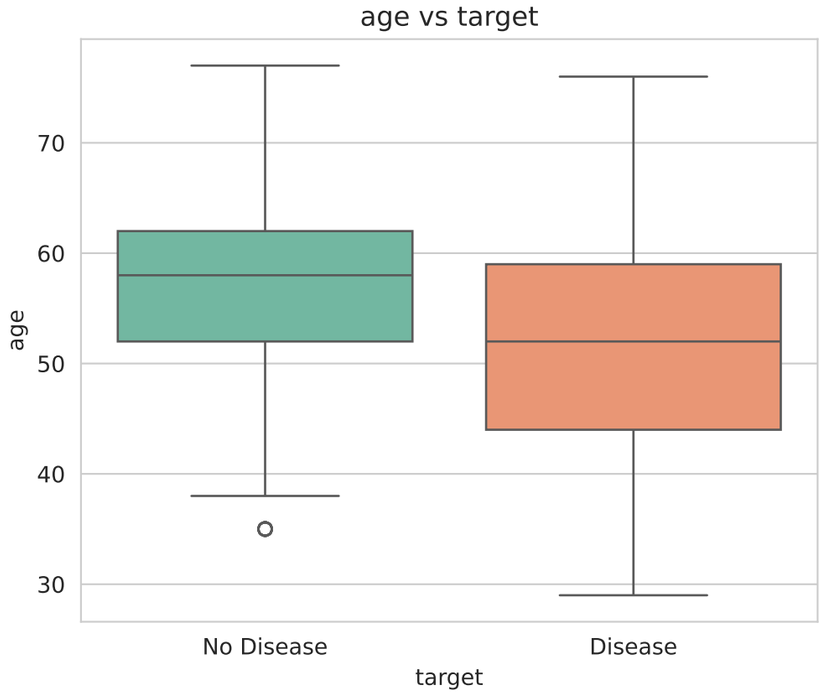

Count of restecg
count
restecg
0
1
2

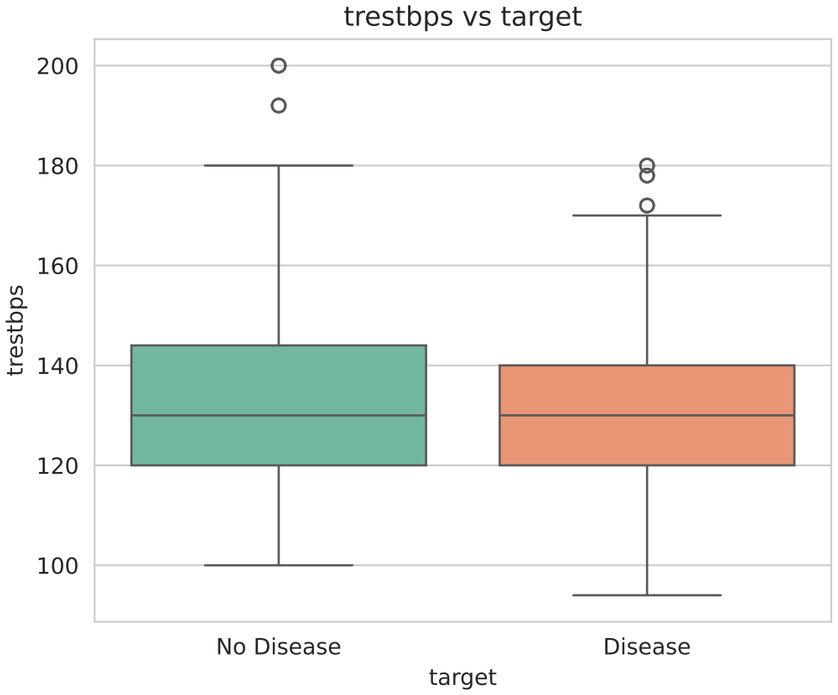

Count of exang
count
exang
0
1

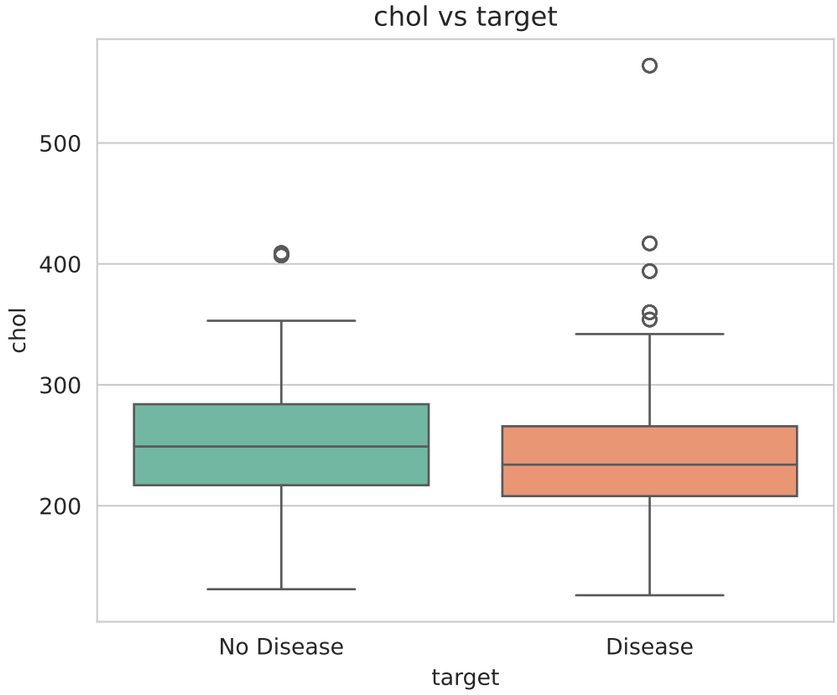

Count of slope
count
slope
0
1
2

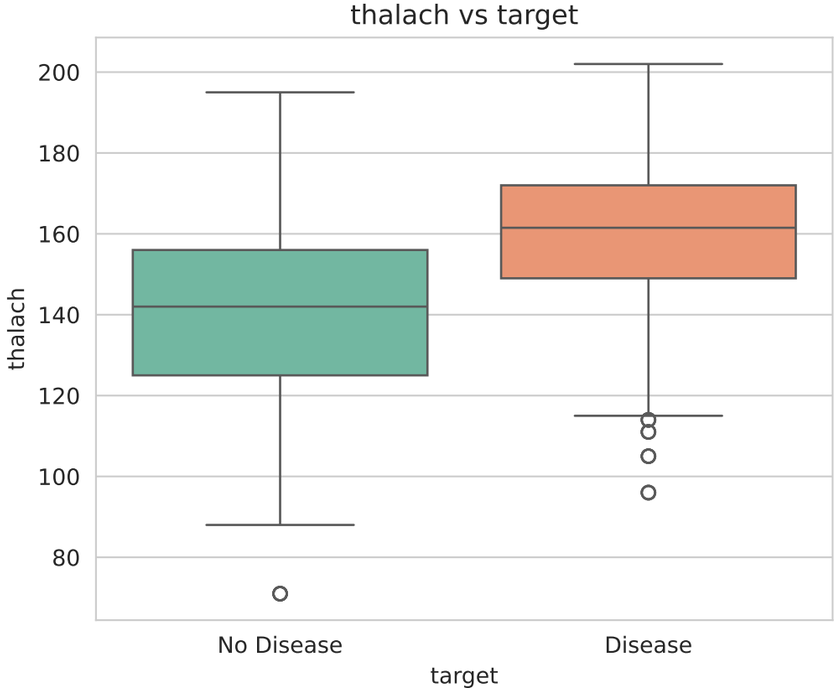

Count of ca
count
ca
0
1
2
3
4

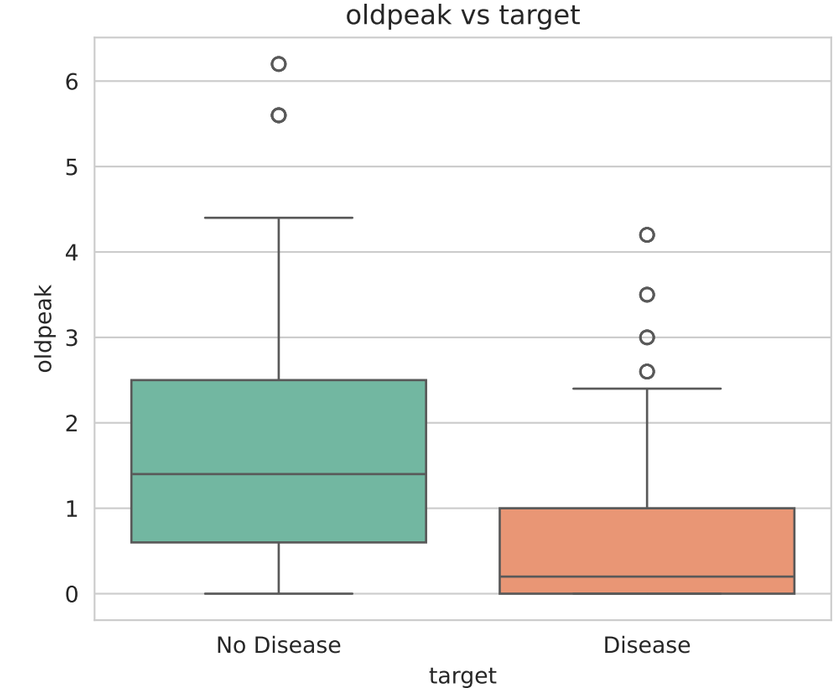

Count of thal
count
thal
0
1
2
3

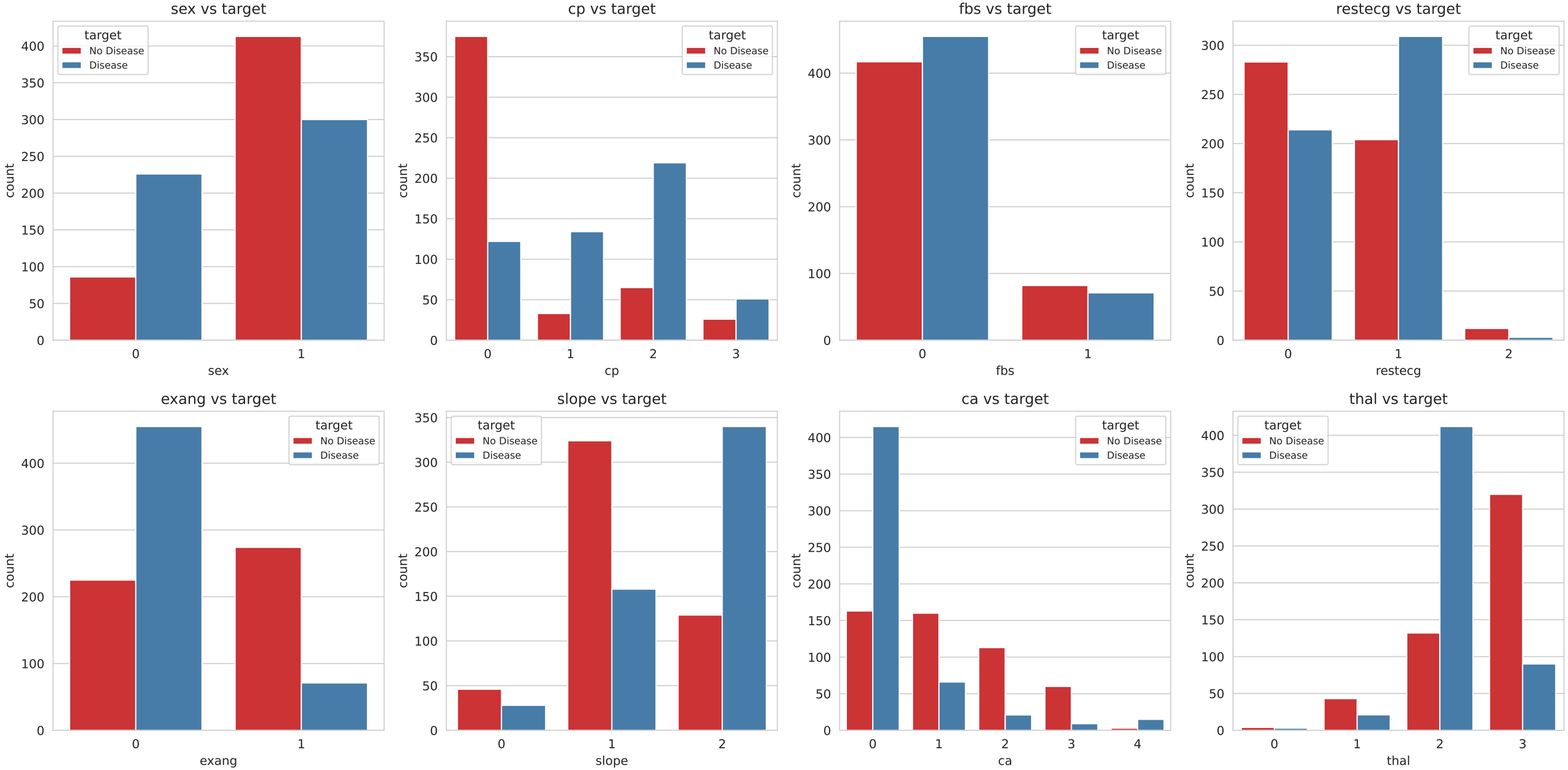

age vs target
age
70
60
50
40
30
No Disease
Disease
target

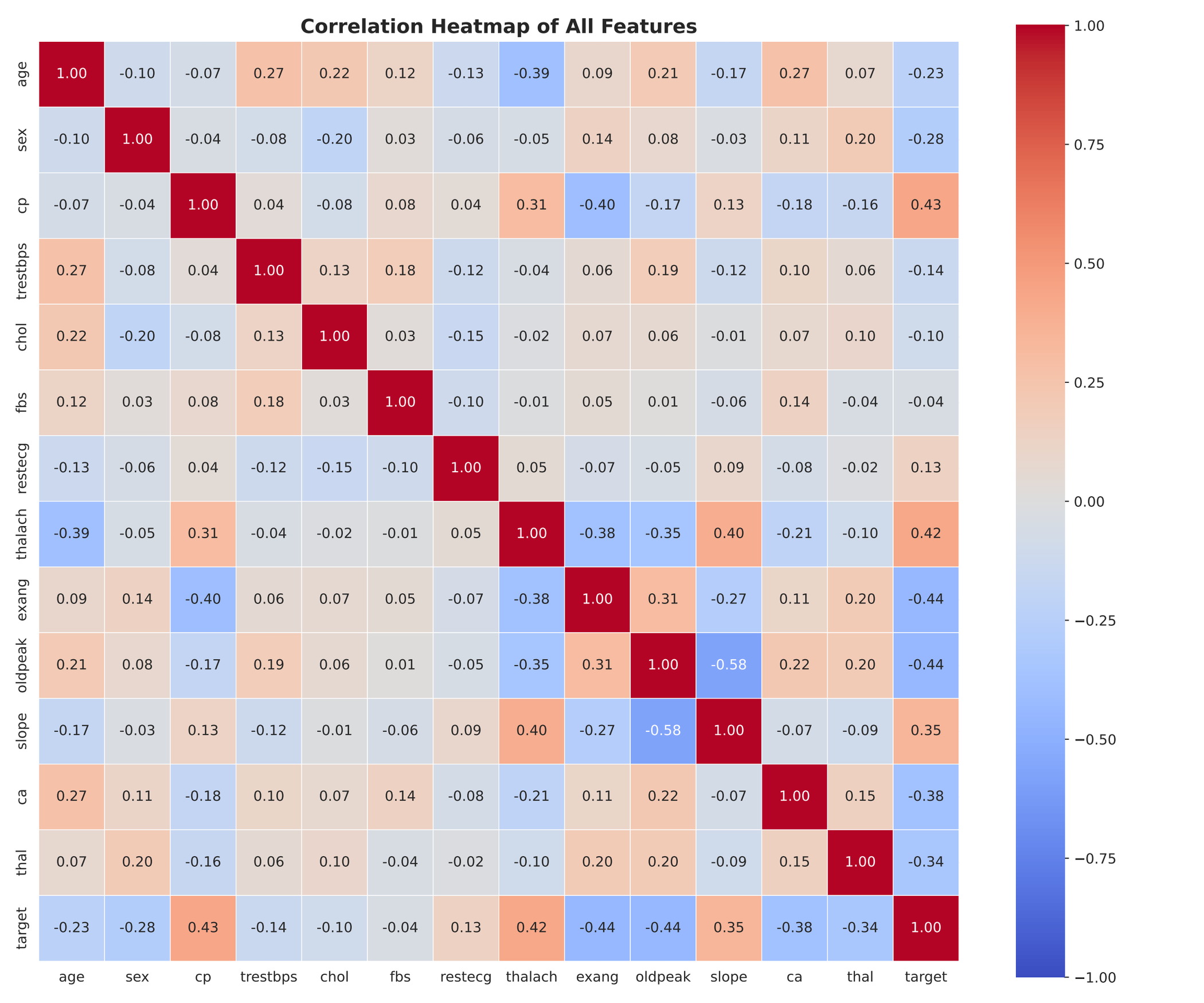

trestbps vs target
trestbps
200
180
160
140
120
100
No Disease
Disease
target

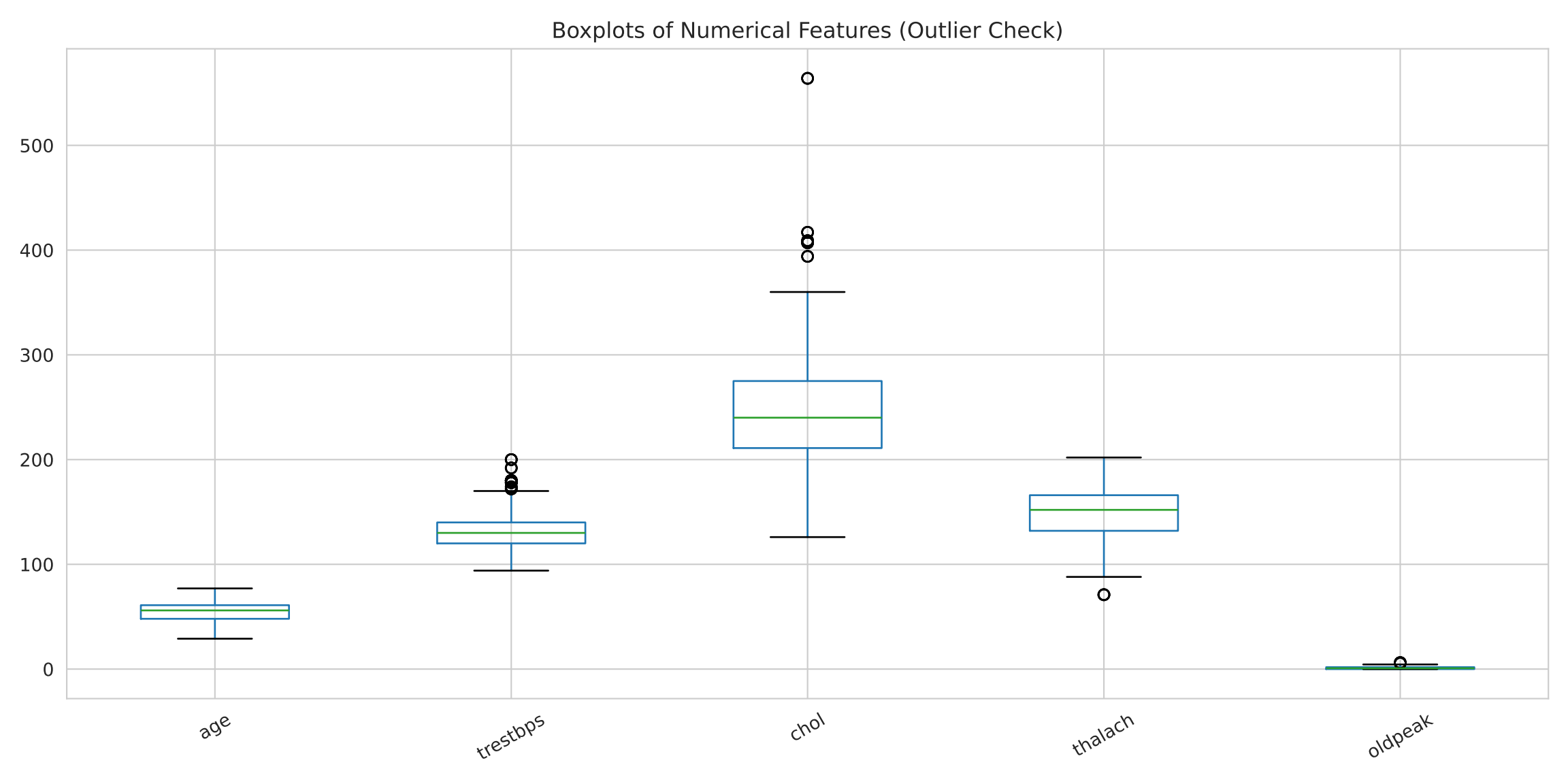

chol vs target
chol
500
400
300
200
No Disease
Disease
target


thalach vs target
thalach
200
180
160
140
120
100
80
No Disease
Disease
target


oldpeak vs target
oldpeak
6
5
4
3
2
1
0
No Disease
Disease
target


sex vs target
target
No Disease
Disease
count
sex
cp vs target
cp
fbs vs target
fbs
restecg vs target
restecg
exang vs target
exang
slope vs target
slope
ca vs target
ca
thal vs target
thal


Correlation Heatmap of All Features
age 1.00 -0.10 -0.07 0.27 0.22 0.12 -0.13 -0.39 0.09 0.21 -0.17 0.27 0.07 -0.23
sex -0.10 1.00 -0.04 -0.08 -0.20 0.03 -0.06 -0.05 0.14 0.08 -0.03 0.11 0.20 -0.28
cp -0.07 -0.04 1.00 0.04 -0.08 0.08 0.04 0.31 -0.40 -0.17 0.13 -0.18 -0.16 0.43
trestbps 0.27 -0.08 0.04 1.00 0.13 0.18 -0.12 -0.04 0.06 0.19 -0.12 0.10 0.06 -0.14
chol 0.22 -0.20 -0.08 0.13 1.00 0.03 -0.15 -0.02 0.07 0.06 -0.01 0.07 0.10 -0.10
fbs 0.12 0.03 0.08 0.18 0.03 1.00 -0.10 -0.01 0.05 0.01 -0.06 0.14 -0.04 -0.04
restecg -0.13 -0.06 0.04 -0.12 -0.15 -0.10 1.00 0.05 -0.07 -0.05 0.09 -0.08 -0.02 0.13
thalach -0.39 -0.05 0.31 -0.04 -0.02 -0.01 0.05 1.00 -0.38 -0.35 0.40 -0.21 -0.10 0.42
exang 0.09 0.14 -0.40 0.06 0.07 0.05 -0.07 -0.38 1.00 0.31 -0.27 0.11 0.20 -0.44
oldpeak 0.21 0.08 -0.17 0.19 0.06 0.01 -0.05 -0.35 0.31 1.00 -0.58 0.22 0.20 -0.44
slope -0.17 -0.03 0.13 -0.12 -0.01 -0.06 0.09 0.40 -0.27 -0.58 1.00 -0.07 -0.09 0.35
ca 0.27 0.11 -0.18 0.10 0.07 0.14 -0.08 -0.21 0.11 0.22 -0.07 1.00 0.15 -0.38
thal 0.07 0.20 -0.16 0.06 0.10 -0.04 -0.02 -0.10 0.20 0.20 -0.09 0.15 1.00 -0.34
target -0.23 -0.28 0.43 -0.14 -0.10 -0.04 0.13 0.42 -0.44 -0.44 0.35 -0.38 -0.34 1.00
age sex cp trestbps chol fbs restecg thalach exang oldpeak slope ca thal target
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Boxplots of Numerical Features (Outlier Check)
500
400
300
200
100
0
age
trestbps
chol
thalach
oldpeak